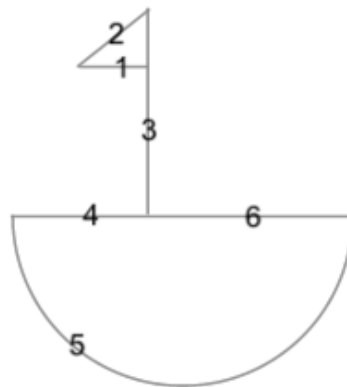


Solution to Tracing Pictograms

David Jiang

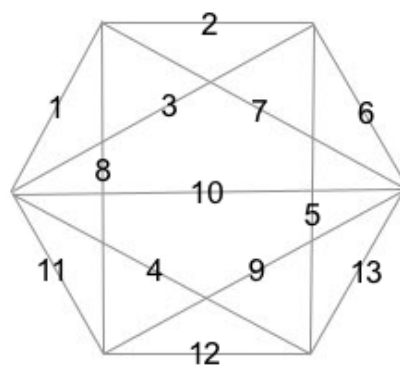
Pictogram #1

You can trace the figure in the following way:



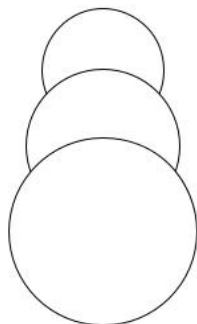
Pictogram #2

You can trace the figure in the following way:



Pictogram #3

The figure is impossible to trace.



Notice that if we convert this into a graph, we get four vertices, each with degree three. When tracing over this figure, we must always enter a vertex from one edge and exit from another. Therefore, each vertex that is not the starting or ending point must have even degree.

Thus, there can be at most two vertices with odd degree. However, in this graph, there are four vertices with odd degree, so we cannot trace the figure. How can the figure be changed so that a solution is possible?